Max Lapin

maksim.lapin@gmail.com | Google Scholar | linkedin.com/in/mlapin

Research Experience

Reka AI, Berlin
Member of Technical Staff — May 2026–Present

Pre-training and world-model research for Rho-1, a 19B model for text, images, and video. Led research on camera-controlled video generation.

Stability AI, Berlin
Research Scientist — Nov 2023–May 2026

Image and video generation, controllability, and world models. Advised Stable Cinemetrics (NeurIPS 2025) and Stable Video-Driven Portraits. Core contributor to Stable Diffusion 3.5 and its ControlNets.

Amazon, Berlin
Senior Applied Scientist (L6), Computer Vision, Robotics AI — Oct 2020–May 2023
Applied Scientist (L5), Computer Vision — Mar 2017–Sep 2020

Led research on RGB-D damage detection and language–image retrieval. Contributed to multimodal product identification; designed the visual-similarity model for Discover Rooms.

Education

Saarland University & Max Planck Institute for Informatics, Saarbrücken, Germany
Ph.D. in Computer Science — 2012–2017
Advisors: Bernt Schiele and Matthias Hein.

Saarland University, Saarbrücken, Germany
M.Sc. in Computer Science — 2009–2012

Selected Publications

Stable Cinemetrics: Structured Taxonomy and Evaluation for Professional Video Generation. A. Chatterjee, R. Entezari, M. Zhuravinskyi, **M. Lapin**, R. Adithyan, A. Raj, C. Baral, Y. Yang, V. Jampani. *NeurIPS 2025*.

Stable Video-Driven Portraits. Mallikarjun B. R., F. Yin, V. Voleti, N. Drobyshev, **M. Lapin**, A. Vasishta, V. Jampani. *arXiv 2025*.

Contrastive Language-Action Pre-training for Temporal Localization. M. Xu, E. Gundogdu, **M. Lapin**, B. Ghanem, M. Donoser, L. Bazzani. *arXiv 2022*.

Analysis and Optimization of Loss Functions for Multiclass, Top-k, and Multilabel Classification. **M. Lapin**, M. Hein, B. Schiele. *IEEE TPAMI, 2018*.

Loss Functions for Top-k Error: Analysis and Insights. **M. Lapin**, M. Hein, B. Schiele. *CVPR 2016*.

Top-k Multiclass SVM. **M. Lapin**, M. Hein, B. Schiele. *NeurIPS 2015 (Spotlight)*.

Efficient Output Kernel Learning for Multiple Tasks. P. Jawanpuria, **M. Lapin**, M. Hein, B. Schiele. *NeurIPS 2015*.

Scalable Multitask Representation Learning for Scene Classification. **M. Lapin**, B. Schiele, M. Hein. *CVPR 2014*.

Learning Using Privileged Information: SVM+ and Weighted SVM. **M. Lapin**, M. Hein, B. Schiele. *Neural Networks 53:95–108, 2014*.

Open Source & Academic Service

libsdca: reference implementation of top-k SVM and top-k entropy losses.

Reviewer: NeurIPS, CVPR, ICCV, ECCV, TPAMI, IJCV, JMLR.

Co-organizer, 10th Amazon Machine Learning Workshop, 2019.

Teaching Assistant, Saarland University: Machine Learning (Fall 2014, Fall 2010), Probabilistic Graphical Models (Fall 2013).